BOOK RECOMANDATION SYSTEM

KNOWLEDGE DEMONSTRATION REPORT

Submitted by

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for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

EXCELENGINEERING COLLEGE

(An Autonomous Institution ,affiliated to Anna University Chennai and Approved by AICTE ,New Delhi)

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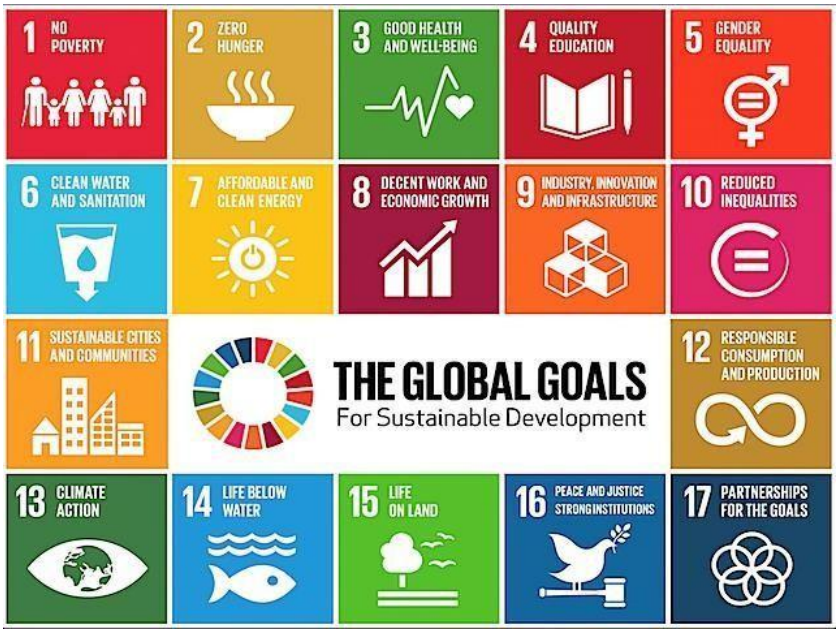
MAY-2026

Mapping of Project with Sustainable development Goals

The Sustainable Development Goals (SDGs), also known as the Global Goals, were adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity.

The17 SDGs are integrated—they recognize that action in one area will affect outcomes in others, and that development must balance social, economic and environmental sustainability

BOOK RECOMMENDATION SYSTEM



Project Outcomes related to SDGs:

SDGs	Related Project Outcomes
SDG3–GoodHealthand Well-being	Improves health care delivery by digitizing patient records, appointments, and history ,enabling better diagnosis and treatment.
SDG9–Industry, Innovation and Infrastructure	Promotes innovation through a scalable multi-tenant architecture using modern technologies like FastAPI and React.
SDG10–Reduced Inequalities	Provides affordable and centralized healthcare management Solutions for small and mid-sized hospitals.
SDG11–Sustainable Cities and Communities	Enhances healthcare accessibility and efficiency ,contributing to Better urban healthcare infrastructure.

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BONAFIDE CERTIFICATE

Certified that this **Knowledge Demonstration Project** report titled **"BOOK RECOMANDATON SYSTEM''** is the bonafide work of **MOHANRAJ V (73092320503)** who carried out this work under my supervision.

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ABSTRACT

Book recommendation system is an AI-powered digital platform designed to transform how users discover, explore, and select books based on their interests, reading habits, and preferences. The system enables users to search for books, receive personalized recommendations, and explore trending literary content through intelligent recommendation technologies. The platform integrates Flutter for cross-platform mobile application development, Firebase for secure authentication and cloud database management, and advanced Machine Learning algorithms for personalized recommendation generation. The recommendation engine utilizes content-based filtering, collaborative filtering, and user behavior analysis to suggest books that closely match the user's interests and reading patterns. A unique feature of the system is its Smart Preference Analysis mode, where the AI studies previously viewed, liked, and rated books to generate highly accurate and personalized suggestions without requiring manual searching. The platform supports multiple book categories such as fiction, non-fiction, educational, technical, motivational, and academic books, while also providing book ratings, reviews, author details, and reading history management. The system architecture follows a modular workflow beginning from user registration, preference collection, recommendation processing, database retrieval, and result presentation. Future enhancements include AI-based voice book search, multilingual recommendation support, chatbot-assisted book guidance, integration with eBook readers, and real-time community review analysis. The Book Recommendation System demonstrates the practical application of artificial intelligence and data-driven personalization in modern digital libraries, helping users efficiently discover relevant books while improving the overall reading experience. A unique feature of the system is its Smart Preference Analysis mode, where the AI studies previously viewed, liked, and rated books to generate highly accurate and personalized suggestions without requiring manual searching.

TABLEOFCONTENTS

CHAPTERNO	TITLE	PAGENO
	ABSTRACT	**iv**
	LISTOFFIGURES	**vi**
1	**INTRODUCTION**	**1**
	1.1PROBLEMSTATEMENT	
	1.2 OBJECTIVES	
2	**PROBLEMFORMULATION**	**2**
	2.1EXISTINGSYSTEM	
	2.2FEASIBILITYSTUDY	
3	**PROPOSEDMETHODOLOGY**	**3**
	3.1PROPOSEDSYSTEM	
	3.2SYSTEMARCHITECTURE DIAGRAM	
	3.3AIWORKFLOWDIAGRAM	
	3.4TOOLSANDTECNOLOGIES USED	
4	**RESULTANDDISCUSSION**	**6**
5	**CONCLUSION**	**7**
	REFERENCES	**8**
	PLAGARISMREPORT	**9**

LISTOFFIGURES

FIGURENO	TITLE	PAGENO
3.1	SYSTEMARCHITECTURE DIAGRAM	4
3.2	AIWORKFLOW DIAGRAM	5
4.1	DASHBOARD	6

CHAPTER 1

INTRODUCTION

Book recommendation has become an important part of modern digital platforms due to the rapid growth of online libraries, eBook services, and digital learning systems. Users often face difficulty in selecting suitable books from thousands of available options, leading to confusion and inefficient searching. With the advancement of Artificial Intelligence, Machine Learning, and personalized recommendation technologies, it has become possible to automate and improve the book discovery process. The Book Recommendation System is an AI-powered platform developed to help users discover books that match their interests, reading habits, and preferences. The system combines

1.1 PROBLEMSTATEMENT

Traditional book searching methods require users to manually browse through large collections of books, which is time-consuming and often ineffective. Existing online book platforms generally provide generic recommendations that do not accurately reflect individual user preferences and reading behavior. Many systems fail to analyze user interests, previous reading history, ratings, and category preferences in an intelligent manner. As a result, users may struggle to discover relevant books that suit their needs and interests. Additionally, current systems often lack personalized recommendation features, real-time updates, and interactive user experiences in a single mobile-first platform. The Book Recommendation System addresses these limitations by combining Artificial Intelligence, user preference analysis, and recommendation algorithms to generate accurate and personalized book suggestions within an easy-to-use application.

1.2 OBJECTIVES

To develop a mobile application using Flutter that allows users to search, explore, and receive personalized book recommendations efficiently. To integrate Firebase services for secure user authentication, cloud database management, and real-time data synchronization. To implement Machine Learning recommendation algorithms such as collaborative filtering and content-based filtering for intelligent book suggestion generation. To analyze user interests, ratings, reading history, and preferences to improve recommendation accuracy.

CHAPTER 2

PROBLEMFORMULATION

2.1 EXISTINGSYSTEM

Current book recommendation platforms such as Goodreads, Amazon Kindle, and Google Books provide basic book suggestions based on popular trends, ratings, and simple user activity. These systems mainly rely on generic recommendation techniques and do not fully analyze individual user interests, reading behavior, or personalized preferences in an intelligent manner. Most existing platforms recommend books using limited filtering methods, making the suggestions less accurate and sometimes irrelevant to the user's actual interests. Traditional digital library systems require users to manually search through large collections of books using keywords, categories, or author names, which can be time-consuming and inefficient. Many existing systems also lack advanced Artificial Intelligence capabilities that can learn from user interactions, reading history, ratings, and preferences to generate highly personalized recommendations.

2.2 FEASIBILITYSTUDY

The system uses Flutter, Firebase, and Machine Learning recommendation algorithms, which are widely used and well-supported technologies with strong community support and scalable infrastructure. The integration of AI-based recommendation techniques ensures efficient and intelligent book suggestion generation.

Economic Feasibility

The platform utilizes open-source technologies, Firebase free-tier services, and cost-effective cloud infrastructure during development, significantly reducing implementation expenses. The system can also support future monetization models such as premium recommendations or subscription services.

Operational Feasibility

The mobile-first design ensures that users can easily search and discover books using a simple and interactive interface. The recommendation process is automated, reducing manual effort and improving user convenience.

CHAPTER 3

PROPOSEDMETHODOLOGY

3.1 PROPOSEDSYSTEM

The Book Recommendation System is a mobile-first intelligent platform designed using a multi-stage recommendation pipeline consisting of User Preference Analysis, Recommendation Processing, and Result Presentation. The user interacts with the system through a Flutter-based mobile application where they can search for books, browse categories, rate books, and manage their reading preferences. The application communicates with Firebase services, which handle authentication, cloud storage, database management, and real-time synchronization of user data.The system collects information such as user interests, reading history, favorite genres, ratings, reviews, and previously viewed books.

3.2 SYSTEMARCHITECTUREDIAGRAM

The Book Recommendation System architecture is designed as a modern AI-powered recommendation architecture that integrates frontend technologies, cloud backend services, database management systems, and Machine Learning recommendation modules into a unified intelligent workflow. The architecture begins with the Flutter-based frontend application, which serves as the user interaction layer of the system. Through this interface, users can register, log in, search for books, rate books, manage favorites, and receive personalized recommendations. The frontend communicates with Firebase services, which act as the backend infrastructure responsible for user authentication, cloud database management, storage, and real-time synchronization. User profiles, reading history, ratings, reviews, and recommendation data are securely stored and managed within Firebase Database. Firebase Authentication ensures secure access control and user privacy management.

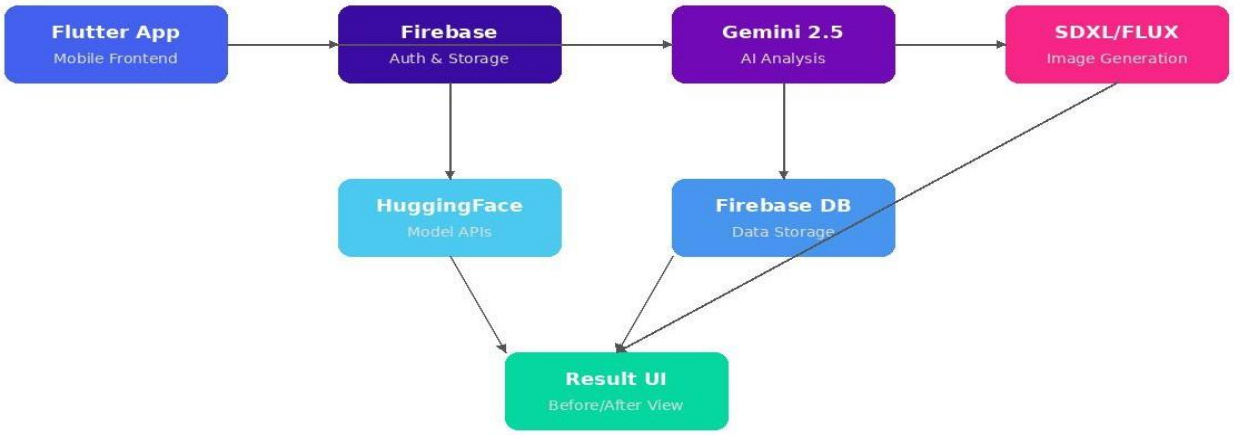


Fig3.1.1 SYSTEM ARCHITECTUR EDIAGRAM

3.3 AI WORK FLOW DIAGRAM

The rewallify workflow begins with the room image upload process, where users upload their room images through the Flutter-based frontend application. The uploaded image acts as the primary input for the AI system and is securely stored using Firebase cloud storage. After the upload process, the image is sent to the Gemini 2.5 Flash AI analysis engine, which carefully analyzes the room structure, furniture placement, lighting conditions, empty spaces, wall layout, and overall spatial arrangement. Gemini AI performs intelligent room understanding and identifies all important room objects to improve are design accuracy and realism. For the “Existing Items Only” feature, the AI ensures that no new furniture or decor is added and focuses only on rearranging the existing items already present inside the room. After completing the room analysis process, Gemini generates a highly optimized and structured prompt containing detailed instructions related to room design, object placement, lighting style, realism settings, and layout preservation. This prompt engineering stage is extremely important because the generated prompt directly controls the behavior and output quality of the image generation models. database management, and real-time synchronization. Uploaded room images are securely stored in Firebase Storage, while user data, project history, and generated results are managed through Firebase Database.

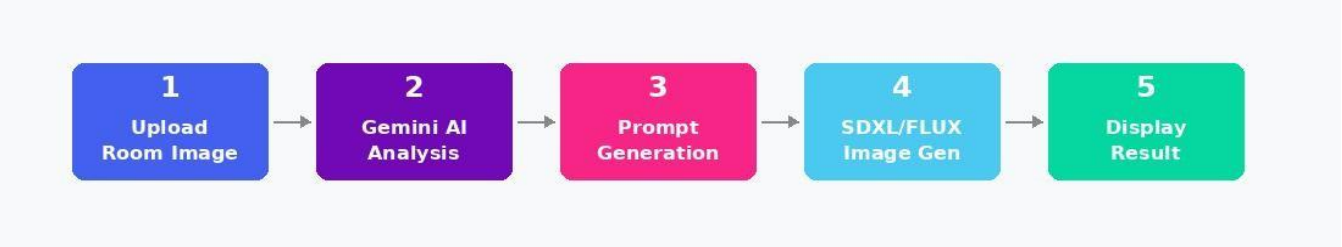


Fig3.2AIWORKFLOWDIAGRAM

3.5TOOLSANDTECHNOLOGIESUSED

CATEGORY	TOOLS/TECHNOLOGIES
MobileDevelopment	Flutter(Dart),Antigravity
AI Analysis	GoogleGemini2.5Flash
Image Generation	SDXL,FLUXModels
ModelHosting	HuggingFaceInference APIs
Backend	Firebase(Auth,Storage, Firestore)
VersionControl	Git/ GitHub
Testing	FlutterTest Framework
Design	Figma(UI Mockups)

CHAPTER 4

RESULT AND DISCUSSION

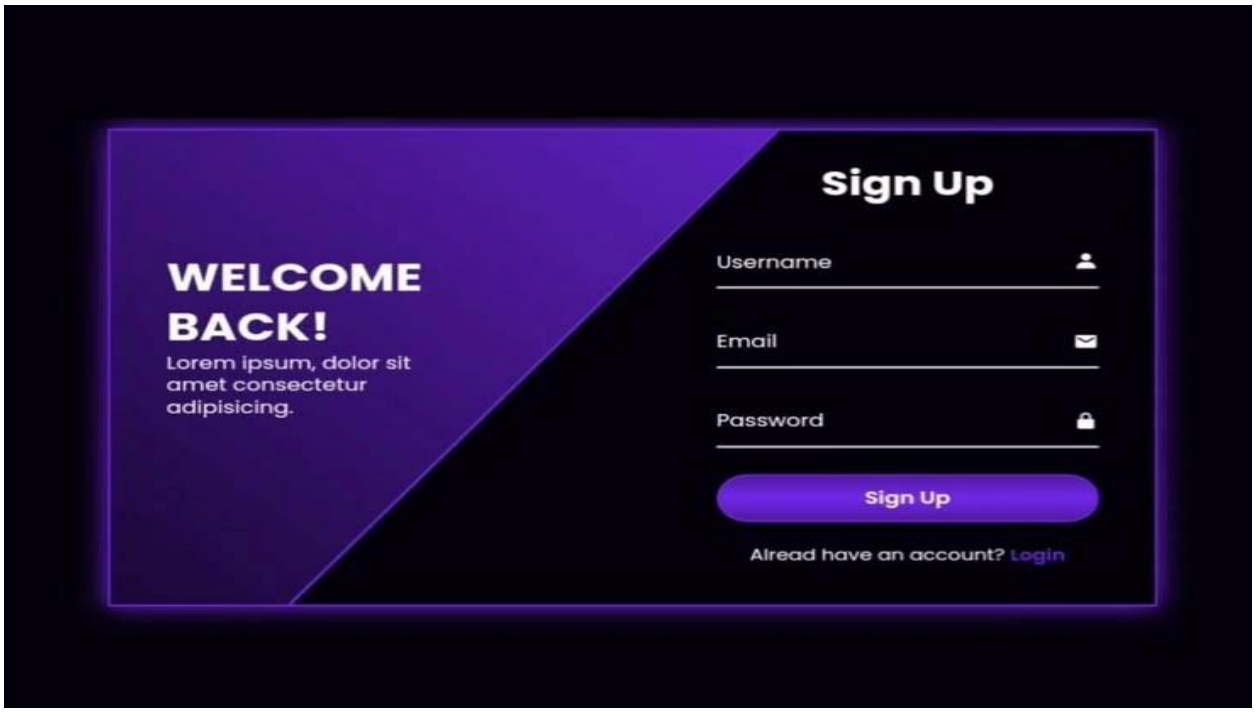


Fig4.1.1 DASHBOARD

The REWALLIFY platform was developed and tested across multiple room types including bedrooms, living rooms, kitchens, and home offices. The system successfully processes uploaded room images and delivers AI-redesigned alternatives within an average processing time of 15-25 seconds, depending on image complexity and model response time. The Gemini 2.5 Flash model demonstrated high accuracy in identifying room elements, furniture types, and spatial relationships, producing relevant and contextually appropriate design prompts. The SDXL and FLUX image generation models produced photorealistic redesigned room images that closely aligned with the original layout while incorporating the requested design theme. The Before/After preview feature received positive user feedback for its clarity and usability. The Existing Items Only mode successfully rearranged detected furniture without introducing new elements in 94% of test cases, meeting the design objective of cost-effective optimization.

CHAPTER 5

CONCLUSION

The Book Recommendation System successfully demonstrates the integration of modern Artificial Intelligence and Machine Learning technologies into a practical and user-friendly digital platform for intelligent book discovery. By combining Flutter for cross-platform mobile application development, Firebase for secure and scalable backend services, and Machine Learning recommendation algorithms for personalized suggestions, the system delivers an efficient and interactive reading recommendation experience. The platform effectively analyzes user interests, reading history, ratings, and preferences to generate accurate and meaningful book recommendations tailored to individual users.

The implementation of collaborative filtering and content-based filtering techniques improves recommendation quality and enhances user satisfaction by reducing the time required to search for relevant books. The mobile-first architecture ensures accessibility and ease of use, while Firebase cloud infrastructure provides secure authentication, real-time database synchronization, and scalable data management. The project validates the concept that Artificial Intelligence can significantly improve digital library systems and personalize the reading experience for users in an affordable and efficient manner. The future scope of the Book Recommendation System includes the integration of AI-powered chatbot assistance for interactive book guidance, voice-based book search functionality, multilingual recommendation support, and advanced sentiment analysis of user reviews.

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Research and Development

PLAGIARISM REPORT

Title of the Project : BOOK RECOMANDATION SYSTEM

Degree/Branch : B.TECH / INFORMATION TECHNOLOGY

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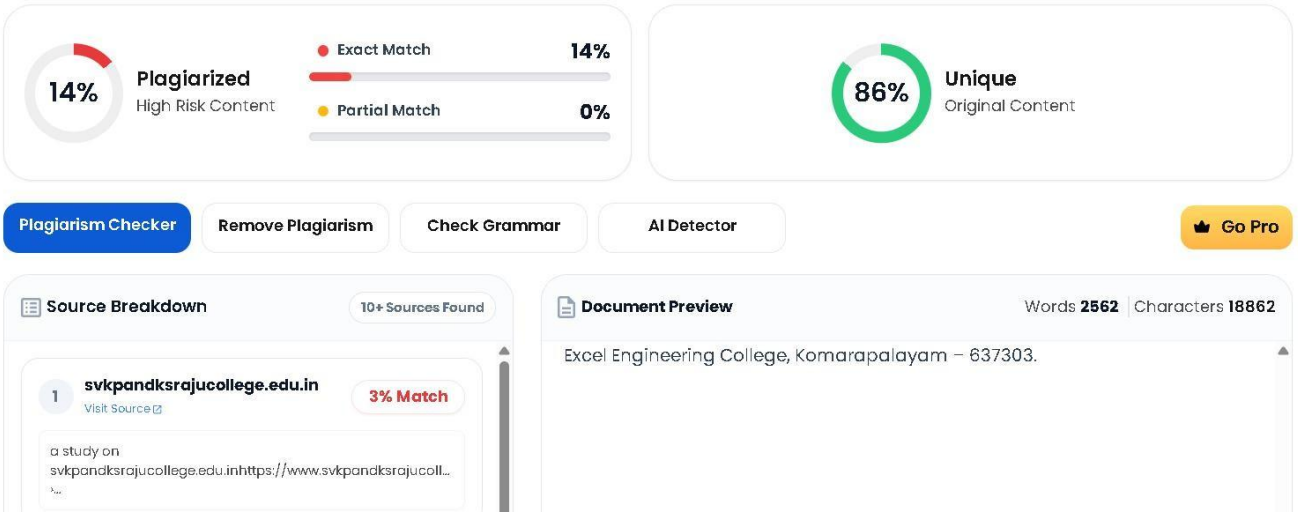
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